THIN ℓ-WEIGHT FILTRATIONS IN THE DRINFELD–KOHNO COMPACT CORE

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Abstract. For finite-dimensional evaluation modules with thin q-characters, the Gautam–Toledano Laredo transfer from quantum loop algebras to Yangians gives the Clebsch–Gordan input needed on the evaluation-prefundamental Drinfeld–Kohno compact core. The tensor product $V(a) \otimes L_i^-(b)$ has an associated graded indexed by the q-character monomials of $V(a)$; if $\chi_q(V) = \sum_m c_m m$, the corresponding prefundamental factor appears with multiplicity c_m. Thin modules give the coefficient-one form. For non-thin fundamental modules, already present in classical types, the correct statement is the multiset-valued filtration, not a multiplicity-free direct sum. The unconditional output is the evaluation-generated compact-core input to MC3; shifted-envelope generation and passage to the full completed Drinfeld–Kohno category are separate completion statements.

1. Introduction

1.1. **The minuscule hypothesis.** The classical Drinfeld–Kohno theorem [5, 18] identifies the monodromy of the Knizhnik–Zamolodchikov connection on conformal blocks for $\widehat{\mathfrak{g}}_k$ with the R-matrix action of the quantum group $U_q(\mathfrak{g})$ at $q = e^{\pi i/(k+h^\vee)}$. Kazhdan and Lusztig extended this to a braided tensor equivalence between the Kazhdan–Lusztig category at level k and the category of type-1 modules over the quantum group [14–17]. Their proof, and the subsequent refinement of Etingof and Kazhdan in terms of quantization of Lie bialgebras [6, 7], leans at several points on the availability of a *minuscule* representation: a fundamental representation whose nonzero weight spaces are all one-dimensional and form a single Weyl group orbit.

The minuscule hypothesis makes the block combinatorics transparent: ordinary weight spaces are one-dimensional and are indexed by a single Weyl orbit. Type A has this property for every fundamental representation. Outside type A it is exceptional. In E_8 it is absent: the adjoint representation is the smallest nontrivial fundamental representation, its nonzero root spaces are one-dimensional, and its zero-weight space has dimension $\mathrm{rk}(E_8) = 8$. A proof depending on minuscule fundamentals therefore cannot be uniform in Lie type.

1.2. **Multiplicity-free ℓ-weights.** q-characters refine ordinary characters by diagonalizing the commuting Drinfeld loop-Cartan generators. The monomials of $\chi_q(L(\Upsilon))$ record the corresponding ℓ-weights. For thin modules, this refinement separates ordinary weight multiplicities into coefficient-one monomials. For non-thin modules, the coefficients of the q-character must be retained.

Theorem 1.1 (Thin fundamental cases; Frenkel–Mukhin, Nakajima, Chari–Moura)**.** *For the fundamental modules whose q-characters are thin,*

$$\chi_q(V_{\omega_i}(a)) = \sum_{s=1}^{d} m_s$$

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with pairwise distinct monomials and coefficient one. Nakajima's quiver-variety computation [20], *the Frenkel–Mukhin algorithm* [11], *and the Chari–Moura formulas* [1] *supply the source computations in their respective ranges. In non-thin fundamental cases the expansion has the form $\chi_q(V) = \sum_m c_m m$ with some $c_m > 1$.*

The categorical input is therefore thinness where it holds, and a multiset-valued q-character expansion where it does not.

1.3. **Statement of the main theorem.** Let $Y\ (\mathfrak{g})$ denote the Yangian of a simple Lie algebra $\mathfrak{g}$ with formal parameter . For a fundamental weight ω_i and spectral parameter $a \in \mathbb{C}$, let $V_{\omega_i}(a)$ be the corresponding fundamental evaluation module, and let $L_i^-(b) \in \mathcal{O}^{\mathrm{sh}}$ denote the negative prefundamental simple in Hernández' shifted category, of highest ℓ-weight indexed by the Drinfeld polynomial of the identity representation with spectral shift b (see Definition 2.3). For a monomial m in the q-character, write $L_i^-(b; m)$ for the simple object whose highest ℓ-weight is obtained from that of $L_i^-(b)$ by multiplication by m.

Theorem 1.2 (Main theorem)**.** *For every finite-dimensional simple Lie algebra $\mathfrak{g}$, every fundamental weight ω_i, and all generic spectral parameters $a, b \in \mathbb{C}$, the tensor product $V_{\omega_i}(a) \otimes L_i^-(b)$ has a finite filtration in $\mathcal{O}^{\mathrm{sh}}$ with associated graded*

$$\operatorname{gr}\bigl(V_{\omega_i}(a) \otimes L_i^-(b)\bigr) \;\cong\; \bigoplus_{m \in \operatorname{Supp}\chi_q(V_{\omega_i}(a))} L_i^-(b; m)^{\oplus c_m}, \qquad \chi_q(V_{\omega_i}(a)) = \sum_m c_m m.$$

The sum is taken with the q-character coefficients. Thin fundamental modules give the coefficient-one form; non-thin fundamental modules retain the displayed multiplicities. If the relevant Ext^1 *groups between all listed simple factors, including repeated factors, vanish on the chosen generic locus, the filtration splits and (1.2) is a Clebsch–Gordan direct sum. Thus the evaluation-prefundamental compact core of* $\mathrm{DK}_{\mathfrak{g}}$ *has a finite filtered Clebsch–Gordan input in every simple Lie type. Completion beyond that core is a separate theorem or hypothesis.*

The restriction to generic spectral parameters excludes the finite union of values of $a - b$ at which extra Hernández A-factor extensions occur. The displayed formula is the associated-graded statement; a direct sum is exactly the additional Ext-vanishing condition in Theorem 1.2.

Theorem 1.2 replaces the minuscule hypothesis of Kazhdan–Lusztig and of earlier type-A treatments of the categorical Clebsch–Gordan filtration (see [2, §11] and the Francis–Gaitsgory pro-nilpotent completion of factorization categories [9]). The minuscule hypothesis is a computational shortcut: it forces classical weight multiplicities to be trivial. The q-character is the correct invariant for the shifted category. Where it is thin, the graded factors occur once; where it is not, the coefficients record the repeated prefundamental factors.

1.4. **Compact-core MC3 input.** The chiral bar complex $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ of a vertex algebra $\mathcal{A}$ admits a Maurer–Cartan element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$ whose degree-2 projection is the classical r-matrix and whose higher-degree projections form the shadow tower; see [19, Vol. I]. The third master claim (MC3) in modular Koszul duality asserts that the degree-3 component of $\Theta_{\mathcal{A}}$, when restricted to the evaluation-generated core, determines all higher-degree components through thick generation and pro-nilpotent completion of the ambient factorization category. Theorem 1.2 removes the Lie-type obstruction on the evaluation-prefundamental core: for every simple $\mathfrak{g}$, the categorical Clebsch–Gordan step needed by the Yangian target is a finite filtered statement whose associated graded is controlled by the q-character coefficients. The claim that the degree-3 shadow projection determines all higher projections belongs to the compact-core MC3 theorem together with its completion hypotheses, not to the representation-theoretic input alone.

1.5. **Minuscule weights and ℓ-weight separation.** Kazhdan and Lusztig [14] treat the equivalence with quantum group modules type by type, with the minuscule weights of the classical types doing the combinatorial work. Etingof and Kazhdan [6, 7] replace this by a universal quantization functor on Lie bialgebras; their proof is type-uniform, but the passage to the Yangian or quantum affine setting reintroduces the minuscule hypothesis through the prefundamental Clebsch–Gordan decomposition of Chari–Pressley [2]. Hernández and Jimbo [13] introduced the shifted category $\mathcal{O}^{\mathrm{sh}}$ and proved a block criterion which is the combinatorial engine for the filtration. Francis and Gaitsgory [9] developed the pro-nilpotent completion formalism that controls the separate passage from a compact-core comparison to the full shifted category.

Theorem 1.1 is the replacement for the minuscule hypothesis in the proof of the filtered categorical Clebsch–Gordan statement. In the language of the Lorgat monograph [19], it supplies the categorical CG input of MC3 for every simple Lie type, including E_8, in which no minuscule representation exists.

2. Preliminaries

2.1. **Lie bialgebras and Yangians.** Let $\mathfrak{g}$ be a finite-dimensional simple Lie algebra over $\mathbb{C}$ with Cartan subalgebra $\mathfrak{h}$, root system Φ, simple roots $\alpha_1, \dots, \alpha_\ell$, fundamental weights $\omega_1, \dots, \omega_\ell$, Weyl group W, dual Coxeter number $h^\vee$, and invariant symmetric form $(\cdot, \cdot)$ normalized so that long roots have squared length 2. The polynomial loop algebra $\mathfrak{g}[t]$ carries a canonical rational Lie bialgebra structure whose cobracket $\delta : \mathfrak{g}[t] \to \mathfrak{g}[t] \otimes \mathfrak{g}[t]$ is determined by the level-stripped rational r-matrix

$$r_{\mathrm{rat}}(u) = \frac{\Omega}{u}, \qquad \Omega = \sum_a I_a \otimes I^a,$$

where $\{I_a\}$ is a basis of $\mathfrak{g}$ and $\{I^a\}$ its dual with respect to $(\cdot, \cdot)$. The affine collision calculation uses the level-dependent trace-form residue $r_k^{\mathrm{coll}}(z) = k\Omega_{\mathrm{tr}}/z$; the Yangian and KZ rational structure use (2.1) with the usual coupling normalization. The two normalizations must not be identified. At affine level $k = 0$ the trace-form double-pole collision residue vanishes, while the simple-pole Lie bracket channel and the level-stripped rational Yangian structure remain nonzero; see [19].

Remark 2.1 (Level normalization). The symbol k belongs to the affine Kac–Moody collision residue. The Yangian rational r-matrix is level-stripped. Thus $k = 0$ is not an abelianization of $\mathfrak{g}[t]$ and does not annihilate the Yangian Maurer–Cartan structure.

Drinfeld [3, 4] quantized $\mathfrak{g}[t]$ to the Yangian $Y(\mathfrak{g}) = Y(\mathfrak{g})$: a Hopf algebra deforming $U(\mathfrak{g}[t])$ along the Lie bialgebra structure (2.1). The Yangian is graded by conformal weight (the eigenvalues of a derivation lifting the loop rotation $t \mapsto t + c$) and admits an evaluation homomorphism $\mathrm{ev}_a : Y(\mathfrak{g}) \to U(\mathfrak{g})$ for each $a \in \mathbb{C}$ in type A, and more subtly in other types.

2.2. **The Drinfeld–Kohno category.**

Definition 2.2 (Drinfeld–Kohno categories)**.** The finite Drinfeld–Kohno category $\mathrm{DK}_{\mathfrak{g}}^{\mathrm{fd}}$ is the braided monoidal category of finite-dimensional $Y(\mathfrak{g})$-modules, with braiding induced by the rational R-matrix attached to (2.1). The shifted category $\mathcal{O}^{\mathrm{sh}}$ of Hernández and Jimbo [13] is larger: it contains the infinite-dimensional prefundamental modules and has locally finite Cartan-loop action with weights in a shifted dominant cone. In this paper $\mathrm{DK}_{\mathfrak{g}}^{\mathrm{ec}}$ denotes the evaluation-prefundamental compact core inside the completed shifted category, generated by fundamental evaluation modules and negative prefundamental simples.

Definition 2.3 (Fundamental and prefundamental objects)**.** For each fundamental weight ω_i and spectral parameter $a \in \mathbb{C}$, the *fundamental evaluation module* $V_{\omega_i}(a)$ is the finite-dimensional simple

$Y(\mathfrak{g})$-module whose Drinfeld polynomial at node i is $1 - az$ and is trivial at all other nodes. The *negative prefundamental module* $L_i^-(b)$ is the simple infinite-dimensional object in $\mathcal{O}^{\mathrm{sh}}$ whose highest ℓ-weight is the "minus-delta at node i with shift b" configuration of [13].

2.3. **The chromatic filtration.** The category $\mathcal{O}^{\mathrm{sh}}$ carries a decreasing *chromatic* or *conformal-weight* filtration $\mathcal{F}^{\geq N} \subset \mathcal{O}^{\mathrm{sh}}$, where $\mathcal{F}^{\geq N}$ consists of objects whose generalized conformal-weight generalized eigenvalues are bounded below by N. The associated graded pieces $\mathcal{F}^{\geq N}/\mathcal{F}^{\geq N+1}$ are finite-dimensional strata in the sense that, at each fixed highest ℓ-weight, the space of conformal-weight-N vectors is finite-dimensional. The compact evaluation-prefundamental core consists of the objects supported on finitely many such strata and built from fundamental evaluation modules and negative prefundamental simples by finite extensions, shifts, cones, and retracts. Identifying this core with the whole compact subcategory of the completed Drinfeld–Kohno category is a separate completion theorem or hypothesis; it is not part of the Clebsch–Gordan statement proved here.

2.4. **ℓ-weights and the multiplicity-free property.** Let $U_q(\hat{\mathfrak{g}})$ denote the quantum affine algebra at the corresponding parameter q. Its commutative Cartan-type subalgebra is generated by the loop elements $\phi_{i,\pm}(z)$ of Drinfeld's second realization, and any finite-dimensional simple $U_q(\hat{\mathfrak{g}})$-module decomposes as a direct sum of joint generalized eigenspaces for these loop generators.

Definition 2.4 (ℓ-weights). An *ℓ-weight* of $U_q(\hat{\mathfrak{g}})$ is a pair $(\psi^+(z), \psi^-(z))$ of ℓ-tuples of formal power series in $z^{\pm 1}$ subject to the compatibility and rationality conditions of Frenkel and Reshetikhin [10]. The *q-character* of a finite-dimensional simple module V is

$$\chi_q(V) \;=\; \sum_m \dim V_m \cdot m,$$

where the sum ranges over ℓ-weights m and V_m is the corresponding joint generalized eigenspace. A module V is *multiplicity-free in ℓ-weights* if $\dim V_m \leq 1$ for every ℓ-weight m.

The central structural data we import from the literature is the finite q-character with its coefficients.

Theorem 2.5 (q-character coefficient data). *For every finite-dimensional simple Lie algebra $\mathfrak{g}$, every fundamental weight ω_i, and every spectral parameter a, the fundamental evaluation module $V_{\omega_i}(a)$ of $U_q(\hat{\mathfrak{g}})$ has a finite q-character*

$$\chi_q(V_{\omega_i}(a)) \;=\; \sum_m c_m m, \qquad c_m = \dim V_m.$$

In the thin cases all coefficients are 1*. In the non-thin cases the coefficients are retained and become multiplicities of the prefundamental graded factors. Frenkel–Reshetikhin* [10] *define the q-character, Nakajima's quiver-variety computation* [20] *supplies the simply-laced source calculation, and Chari–Moura* [1] *give explicit formulas for the classical families in their range.*

2.5. **Transfer between Yangians and quantum affine algebras.** The bridge between the Yangian $Y(\mathfrak{g})$ and the quantum affine algebra $U_q(\hat{\mathfrak{g}})$ is the theorem of Gautam and Toledano Laredo:

Theorem 2.6 (Gautam–Toledano Laredo, [12]). *After choosing the logarithmic branch and avoiding the singular hyperplanes of the parameters, there is a meromorphic tensor equivalence*

$$\Phi_{\mathrm{GTL}} \;:\; \mathrm{Rep}^{\mathrm{fd}}(U_q(\hat{\mathfrak{g}})) \xrightarrow{\sim} \mathrm{Rep}^{\mathrm{fd}}(Y(\mathfrak{g}))$$

between the corresponding finite-dimensional module categories. It sends simples to simples and identifies the Drinfeld rational fraction data on both sides. In particular, Φ_{GTL} preserves the q-character coefficients.

Consequently, Theorem 2.5 transfers to the Yangian as coefficient data: the fundamental evaluation module $V_{\omega_i}(a)$ of $Y(\mathfrak{g})$ has the same finite multiset of ℓ-weights. Thin modules remain coefficient-one; non-thin modules retain their multiplicities. The Hernández–Jimbo block criterion used below is combinatorial in the Drinfeld rational fractions and therefore transfers on the same branch.

3. Statement of the main theorem

Theorem 3.1 (Filtered categorical Clebsch–Gordan, all types)**.** *Let $\mathfrak{g}$ be any finite-dimensional simple Lie algebra over $\mathbb{C}$, let ω_i be any fundamental weight, and let $a, b \in \mathbb{C}$ be spectral parameters lying in the finite-filtration locus of $\mathcal{O}^{\mathrm{sh}}$. Let*

$$\chi_q(V_{\omega_i}(a)) = \sum_m c_m m,$$

and write $L_i^-(b; m)$ for the negative prefundamental simple obtained from $L_i^-(b)$ by multiplying its highest ℓ-weight by m. Then:

(1) *The tensor product $V_{\omega_i}(a) \otimes L_i^-(b)$ has a finite filtration in $\mathcal{O}^{\mathrm{sh}}$ whose associated graded is*

$$\mathrm{gr}\big(V_{\omega_i}(a) \otimes L_i^-(b)\big) \;\cong \bigoplus_{m \in \mathrm{Supp}\,\chi_q(V_{\omega_i}(a))} L_i^-(b; m)^{\oplus c_m}.$$

(2) *The number of graded factors counted with multiplicity is $\sum_m c_m = \dim V_{\omega_i}$. Thin modules are precisely the coefficient-one case.*

(3) *If $\mathrm{Ext}^1_{\mathcal{O}^{\mathrm{sh}}}(S, S') = 0$ for all simple factors S, S' occurring in the multiset*

$$\{\, L_i^-(b; m) \text{ repeated } c_m \text{ times} \,\},$$

then the filtration splits and ((1)) is a direct-sum Clebsch–Gordan decomposition.

(4) *The object $V_{\omega_i}(a) \otimes L_i^-(b)$ belongs to the evaluation-prefundamental compact core generated by fundamental evaluation modules and negative prefundamental simples. Identifying that core with the full compact subcategory of a completed Drinfeld–Kohno category is a separate completion theorem or hypothesis.*

Remark 3.2 (Scope of "generic"). The generic locus excluded in the statement is the union of complex hypersurfaces on which additional Hernández A-factor extensions appear. On the finite-filtration locus the associated graded formula is the theorem. On the semisimple sublocus cut out by the Ext-vanishing condition in Theorem 3.1(3), the filtration splits.

4. Proof of the main theorem

The argument separates the character identity, the Yangian transfer, the highest-ℓ-weight filtration, and the Ext-vanishing splitting criterion.

4.1. **Character identity.**

Lemma 4.1 (q-character identity, all types)**.** *For every simple Lie algebra $\mathfrak{g}$, every fundamental weight ω_i, and all spectral parameters a, b:*

$$\big[V_{\omega_i}(a) \otimes L_i^-(b)\big] \;= \sum_{m \in \mathrm{Supp}\,\chi_q(V_{\omega_i}(a))} c_m\, [L_i^-(b; m)],$$

in the completed Grothendieck group of the shifted category, where $\chi_q(V_{\omega_i}(a)) = \sum_m c_m m$.

Proof. The q-character records the joint generalized eigenspaces of the Drinfeld Cartan currents. Tensoring a highest-ℓ-weight prefundamental object by a finite-dimensional module shifts the highest ℓ-weight by each monomial of the finite q-character, with coefficient equal to the corresponding ℓ-weight dimension. This is the Hernández–Jimbo highest-ℓ-weight calculus in $\mathcal{O}^{\mathrm{sh}}$ [13]. The resulting identity in the completed Grothendieck group is (4.1). □

4.2. q-character coefficients for Yangian fundamentals.

Lemma 4.2 (q-character coefficients for Yangian fundamentals)**.** *For every simple $\mathfrak{g}$ and every fundamental weight ω_i, the Yangian fundamental evaluation module $V_{\omega_i}(a)$ (viewed as a $Y(\mathfrak{g})$-module in $\mathcal{O}^{\mathrm{sh}}$) has the same q-character coefficient multiset as its quantum-loop counterpart on the Gautam–Toledano Laredo branch.*

Proof. For the quantum affine algebra $U_q(\hat{\mathfrak{g}})$, Theorem 2.5 gives the finite coefficient data. For the Yangian $Y(\mathfrak{g})$, apply the Gautam–Toledano Laredo equivalence Φ_{GTL} of Theorem 2.6. This equivalence is exact, sends simples to simples, and identifies the Drinfeld rational fractions governing the commuting Cartan currents. Therefore the joint generalized eigenspace dimensions, equivalently the q-character coefficients, are preserved [12]. □

4.3. Highest-ℓ-weight filtration.

Lemma 4.3 (Filtration by prefundamental factors)**.** *Let $\chi_q(V_{\omega_i}(a)) = \sum_m c_m m$, and let $b \in \mathbb{C}$ be a spectral parameter in the finite-filtration locus. Then $V_{\omega_i}(a) \otimes L_i^-(b)$ admits a finite filtration whose graded factors are the simples $L_i^-(b;m)$ with multiplicity c_m.*

Proof. The shifted category $\mathcal{O}^{\mathrm{sh}}$ is controlled by highest ℓ-weights and the $A_{j,c}$ dominance order of Hernández–Jimbo [13]. Tensoring a prefundamental simple by a finite-dimensional module produces a finite object in this highest-ℓ-weight order. Its possible highest ℓ-weights are obtained by multiplying the highest ℓ-weight of $L_i^-(b)$ by the monomials in $\chi_q(V_{\omega_i}(a))$. Lemma 4.2 gives these monomials and their coefficients on the Yangian side. The standard highest-ℓ-weight filtration in $\mathcal{O}^{\mathrm{sh}}$ therefore has precisely the listed simple graded factors, each repeated according to its q-character coefficient. □

4.4. Splitting criterion.

Lemma 4.4 (Ext-vanishing splits the filtration)**.** *Let M be an object of $\mathcal{O}^{\mathrm{sh}}$ with a finite filtration whose graded factors are $S_1, \ldots, S_d$. If $\mathrm{Ext}^1_{\mathcal{O}^{\mathrm{sh}}}(S_j, S_k) = 0$ for all j, k, then $M \cong \bigoplus_j S_j$.*

Proof. Split the filtration inductively. At each stage the extension class lies in an Ext^1 group between one graded factor and the direct sum of the preceding factors. The stated vanishing kills that class, so the short exact sequence splits. Iteration gives the direct sum. □

Proof of Theorem 3.1. Set $M = V_{\omega_i}(a) \otimes L_i^-(b)$. Then M is an object of $\mathcal{O}^{\mathrm{sh}}$: the tensor factor $V_{\omega_i}(a)$ is finite-dimensional, $L_i^-(b)$ is in $\mathcal{O}^{\mathrm{sh}}$ by [13], and tensoring with a finite-dimensional module preserves $\mathcal{O}^{\mathrm{sh}}$. Lemma 4.3 gives the finite filtration and identifies its graded factors. The number of factors counted with multiplicity is $\sum_m c_m = \dim V_{\omega_i}$ by Lemma 4.2. Lemma 4.4 gives the split Clebsch–Gordan decomposition on the Ext-vanishing sublocus. Finally, the evaluation-prefundamental compact core is by definition the thick idempotent-complete subcategory generated by fundamental evaluation modules and negative prefundamental simples. Since thick subcategories are closed under tensoring with generators, finite extensions, cones, and retracts, the tensor product and its listed graded factors lie in that core. □

5. Compact-core MC3 input

Let $\mathcal{A}$ be a vertex algebra and let $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ denote its chiral bar complex on ordered configuration space. Recall that the bar complex is built on the augmentation ideal $\bar{\mathcal{A}} = \ker(\varepsilon)$, not on $\mathcal{A}$ itself; we write

$$\bar{B}^{\mathrm{ch}}(\mathcal{A}) \;=\; T^c(s^{-1}\bar{\mathcal{A}}) \otimes \mathcal{O}(\overline{C}_\bullet(X)),$$

where $\mathcal{O}(\overline{C}_n(X))$ is the sheaf of rational functions on the Fulton–MacPherson compactification with logarithmic poles along the diagonal divisor. The bar complex carries a Maurer–Cartan element

$$\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$$

whose degree-r projection $\pi_{r,0}(\Theta_{\mathcal{A}})$ is the shadow datum at depth r; see [19] for the detailed construction.

Corollary 5.1 (Compact-core MC3 input for all simple types)**.** *Let $\mathfrak{g}$ be any finite-dimensional simple Lie algebra and let $\mathcal{A}_{\mathfrak{g}} = V_k(\mathfrak{g})$ be the affine vertex algebra at level k away from the critical level $k = -h^\vee$. On the thick closure of fundamental evaluation modules and negative prefundamental simples in* $\mathrm{DK}_{\mathfrak{g}}$*, the finite filtered Clebsch–Gordan input required by the compact-core MC3 argument is available in every simple Lie type. The assertion that the degree-3 projection $\pi_{3,0}(\Theta_{\mathcal{A}_{\mathfrak{g}}})$ determines all higher projections then requires the separate compact-core MC3 theorem and its completion hypotheses.*

Proof. The degree-2 projection $\pi_{2,0}(\Theta_{\mathcal{A}_{\mathfrak{g}}})$ is the Yangian rational r-matrix in the level-stripped normalization (2.1); the affine vertex algebra has collision residue $k\Omega_{\mathrm{tr}}/z$. The two normalizations agree only after the level convention is fixed. The degree-3 projection satisfies the classical Yang–Baxter equation together with its first Massey-product obstruction, but the passage from degree 3 to degree $r \geq 4$ is the separate pro-nilpotent completion problem.

By Theorem 3.1, for every simple Lie type including E_8, the evaluation-prefundamental compact core has the required filtered Clebsch–Gordan input; on the Ext-vanishing semisimple locus this input is a direct-sum decomposition. The lift from Grothendieck-group Baxter TQ relations to derived exact triangles in the completed antidominant shifted envelope is the completion statement of [19, §8]. The Francis–Gaitsgory pro-nilpotent formalism [9] then controls the separate passage from the compact core to the full ind-completion. The representation-theoretic obstruction is the availability of this finite filtered q-character input, and Theorem 3.1 supplies it for all simple types. □

Remark 5.2 (The role of the level)*.* The affine level k enters the collision residue $r^{\mathrm{coll}}_k(z) = k\Omega_{\mathrm{tr}}/z$ and the Sugawara scalar, not the level-stripped Yangian rational r-matrix (2.1). At $k = 0$ the trace-form double-pole residue vanishes, but the simple-pole Lie bracket channel and the Yangian MC structure remain. The critical level $k = -h^\vee$ is instead a singular Feigin–Frenkel centre regime and is excluded from Corollary 5.1.

6. Examples by simple type

The input in Theorem 3.1 is the q-character coefficient multiset. Minuscule fundamentals give coefficient-one formulas. Non-minuscule fundamentals require the actual coefficients of $\chi_q(V_{\omega_i}(a))$.

6.1. **Type A_n.** For $\mathfrak{g} = \mathfrak{sl}_{n+1}$, every fundamental representation is minuscule. The q-character support is coefficient-one, and on the Ext-vanishing locus the associated graded formula becomes the usual split Clebsch–Gordan decomposition indexed by the weights of V_{ω_i}.

6.2. **Classical types B_n, C_n, D_n.** The minuscule fundamentals are the spin representation in type B_n, the defining representation in type C_n, and the vector and two half-spin representations in type D_n. These are the coefficient-one cases. The remaining fundamentals are treated by the Chari–Moura q-character formulas [1]. The associated graded in Theorem 3.1 uses the coefficients c_m from those formulas; it is not an all-types multiplicity-free statement.

6.3. **Exceptional types.** In E_6 the two minuscule fundamentals are coefficient-one. In E_7 the unique minuscule fundamental is coefficient-one. The other exceptional fundamentals, and every fundamental of E_8, F_4, and G_2, are governed by their q-character coefficient data. For E_8, the adjoint fundamental has an ordinary zero weight space of dimension equal to the rank, so ordinary weights cannot supply a multiplicity-free Clebsch–Gordan formula. The filtered statement retains the coefficients of $\chi_q(V_{\omega_i}(a))$ and then applies the prefundamental filtration. A split formula requires the Ext-vanishing hypothesis of Theorem 3.1.

Remark 6.1 (q-character input across simple types). The following table records the input used in Theorem 3.1.

Type	Minuscule count	Key input	Output
A_n	n	minuscule coefficient-one support	filtered core
B_n	1	Chari–Moura coefficients	filtered core
C_n	1	Chari–Moura coefficients	filtered core
D_n	3	Chari–Moura coefficients	filtered core
E_6	2	Nakajima coefficients	filtered core
E_7	1	Nakajima coefficients	filtered core
E_8	0	Nakajima coefficients	filtered core
F_4	0	q-character coefficients	filtered core
G_2	0	q-character coefficients	filtered core

The minuscule hypothesis (all ordinary weight multiplicities trivial) fails for the exceptional types E_8, F_4, G_2 and for the intermediate fundamentals of the classical types. The q-character coefficient multiset is the uniform replacement.

7. Conclusion

The absence of minuscule representations in E_8 is a recurring technical obstruction in the categorical parts of the quantum geometric Langlands programme. Traditional proofs construct equivalences of braided monoidal categories type by type, and E_8 is either omitted or handled by a folding or restriction procedure that reduces the representation theory to a setting with minuscule weights. The coefficient-aware q-character filtration supplies the replacement input on the Drinfeld–Kohno evaluation-prefundamental core.

For the quantum geometric Langlands correspondence of Gaitsgory and collaborators, the categorical CG step appears in the proof that the spectral side (quasi-coherent sheaves on the moduli of local systems) and the automorphic side (D-modules on Bun_G) are related by a functor of factorization categories thickly generated by fundamental evaluation objects. Corollary 5.1 gives the Lie-type input for that construction on the evaluation-prefundamental compact core, including E_8. The existence of the full functor on completed ambient categories remains the separate completion problem.

In modular Koszul duality, this supplies the type-uniform compact-core Clebsch–Gordan input for MC3. The pro-nilpotent passage from compact-core filtrations to the full ambient ind-category is governed by the completion formalism of Francis–Gaitsgory and by the module-category framework of [19].

Remark 7.1 (Limitations). Three caveats accompany Theorem 3.1. First, the generic spectral parameter hypothesis is genuine: on the Hernández A-linked hypersurfaces extra extensions may appear. Second, the theorem proves an associated-graded filtered statement; the direct-sum Clebsch–Gordan formula requires the stated Ext-vanishing. Third, the theorem covers the evaluation-prefundamental compact core; passage to completed categories requires the Francis–Gaitsgory formalism and the module-category framework of [19].

Remark 7.2 (Comparison with prior literature). Drinfeld's "Quasi-Hopf algebras" [5] introduced the Drinfeld associator and the universal monodromy. Kazhdan–Lusztig [14–17] proved the categorical equivalence type by type in positive-level affine categories. Etingof–Kazhdan [6, 7] gave a type-uniform quantization. Hernández and Jimbo [13] introduced the shifted category $\mathcal{O}^{\mathrm{sh}}$ and the block criterion used here. Chari–Pressley [2] proved the type-A categorical CG. Francis–Gaitsgory [9] supplied the factorization pro-nilpotent formalism. The q-character coefficient filtration is the uniform replacement for the minuscule hypothesis, including E_8.

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